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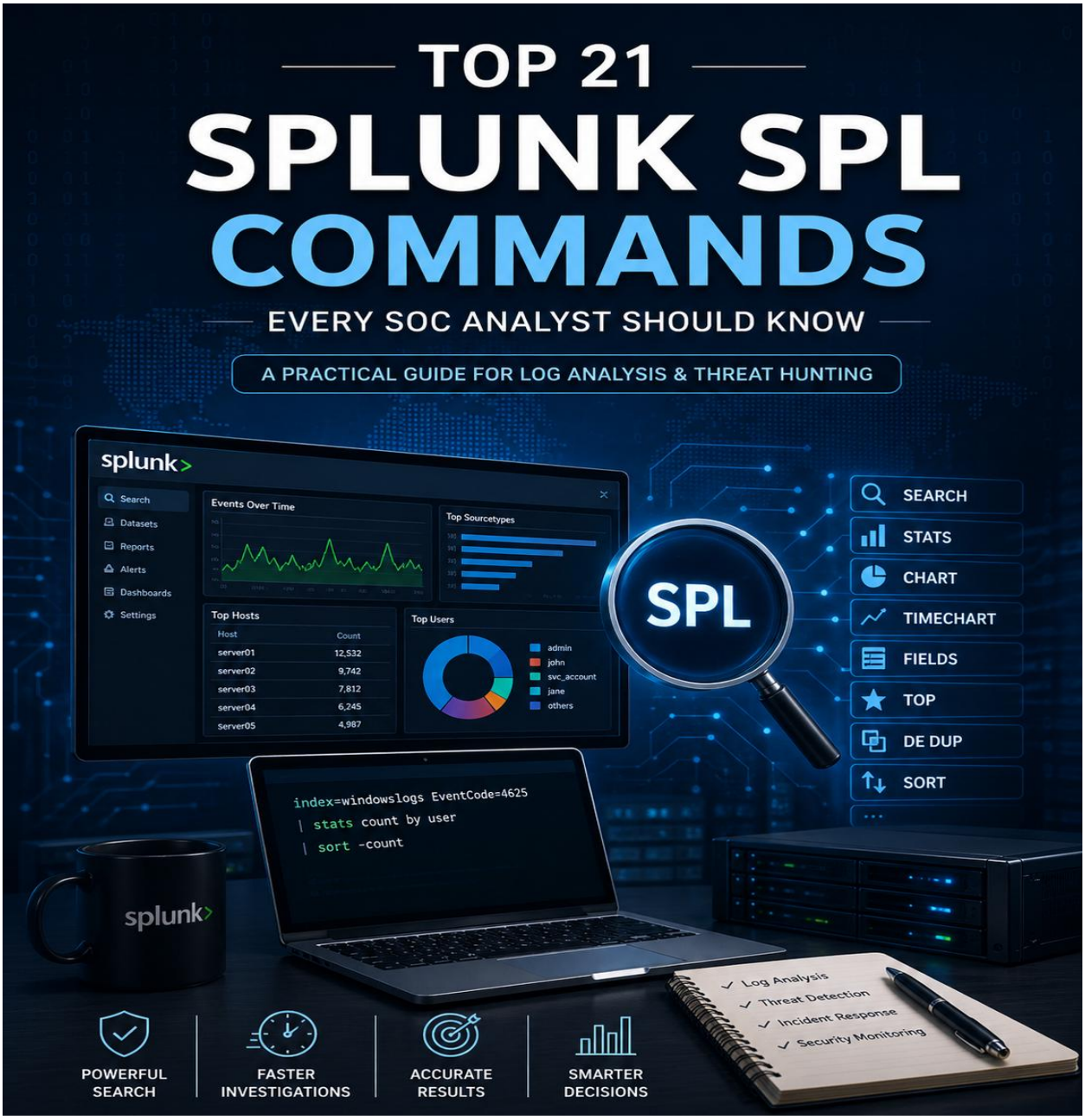

Analysis Top 21 Splunk SPL Commands Every SOC Analyst Should Know

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1. Introduction

Splunk is a widely used Security Information and Event Management (SIEM) platform. It helps security analysts collect, search, visualize, and analyze logs generated from different systems and devices.

To make log searching faster and more efficient, Splunk uses its own query language called Search Processing Language (SPL).

SPL consists of various commands, functions, operators, and arguments that allow analysts to retrieve meaningful data from large volumes of logs quickly.

In this document, I will discuss some important Splunk queries that make log analysis easier, faster, and more efficient.

2. Top 21 SPL Commands

1. Index

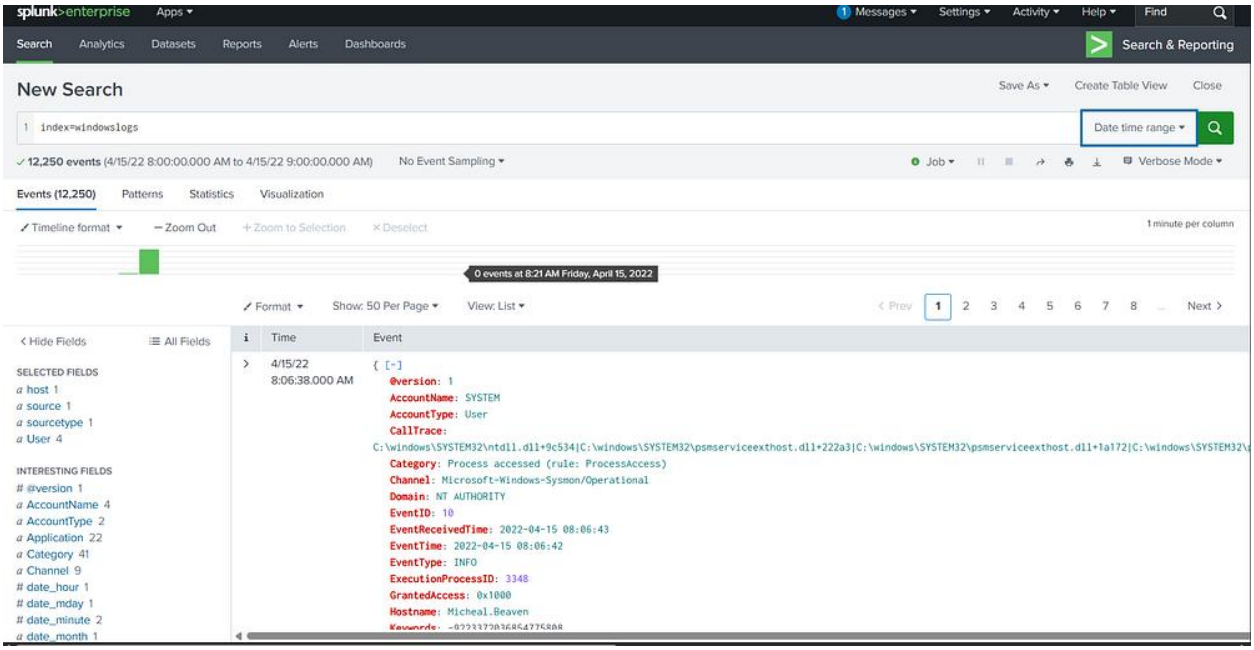
In an organization's IT infrastructure, many devices generate logs every second. These can includes windows system, linux servers, firewalls, network devices and so on. To make searching easier, Splunk stores logs in separate categories called Indexes.

For Example:

- windowslogs: Windows event logs
- packetlogs: Network traffic logs
- serverlogs: Server activity logs

Example Query:

```
index=windowslogs
```



2. Search Operator

Splunk search operators are the building blocks used to contrust any search query. This operators help combine conditions, filter results, compare values, and build smarter searches. They are very important for SOC analysts because they make threat hunting and log analysis faster.

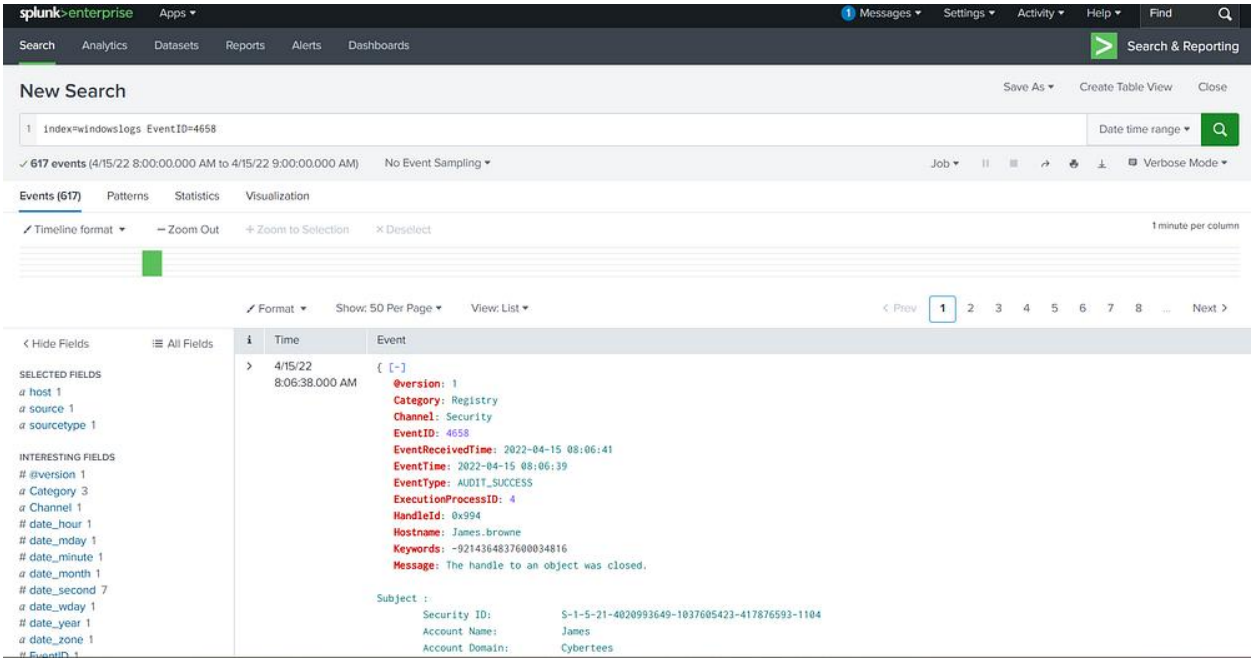
Relational Operators

This operators are used to compare the expression or values.

- = (Equal To)
- != (Not Equal To)
- > (Greater Than)
- < (Less Than)
- >= (Greater Than Or Equal To)
- <= (Less Than Or Equal To)

Example:

```
index=windowslogs EventID=4658
```



Logical Operators:

Logical Operators are used to combine multiple operations.

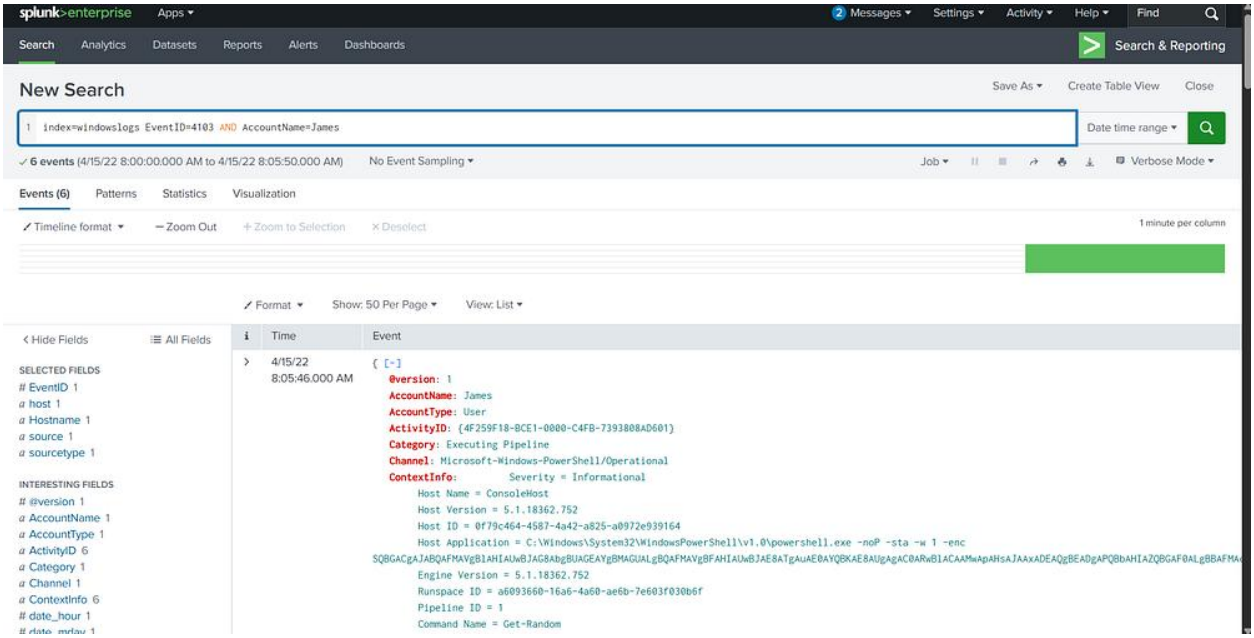
- AND: Results where both conditions are true
- OR: Results where either condition is true
- NOT: Excludes specific results
- IN: To match condition form a long list

Example:

```
index=windowslogs  EventID=4103  AND  AccountName=James
```

```
index=windowslogs EventID=4103 AccountName=James
```

Both the queries are valid



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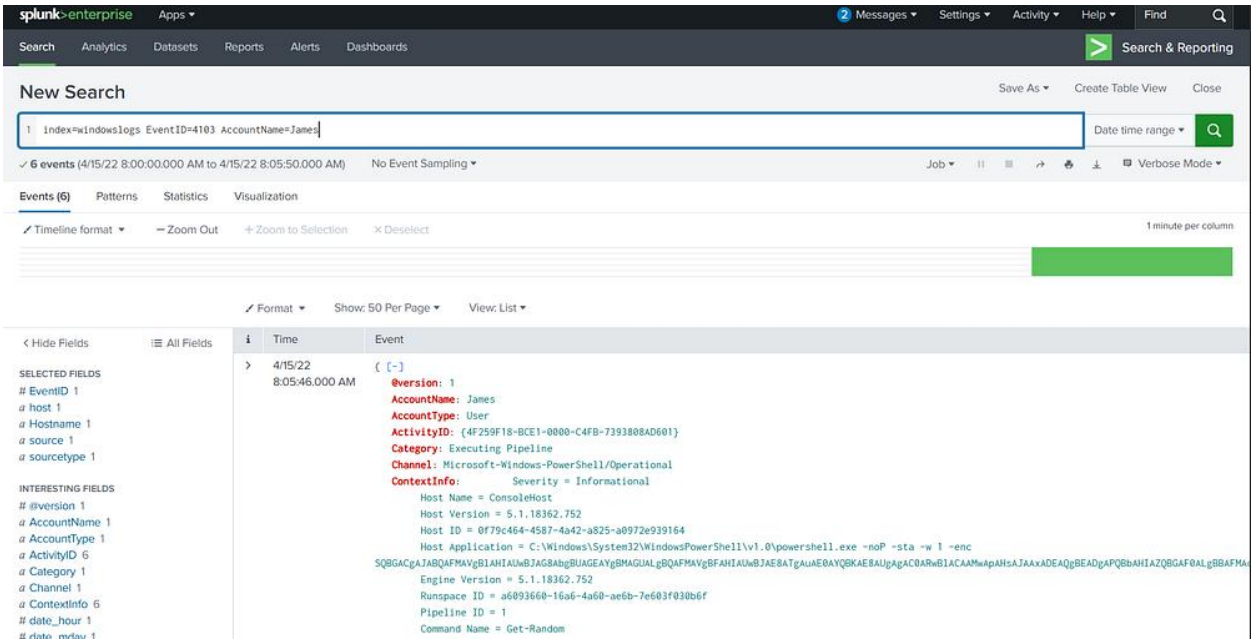


3. Wildcard

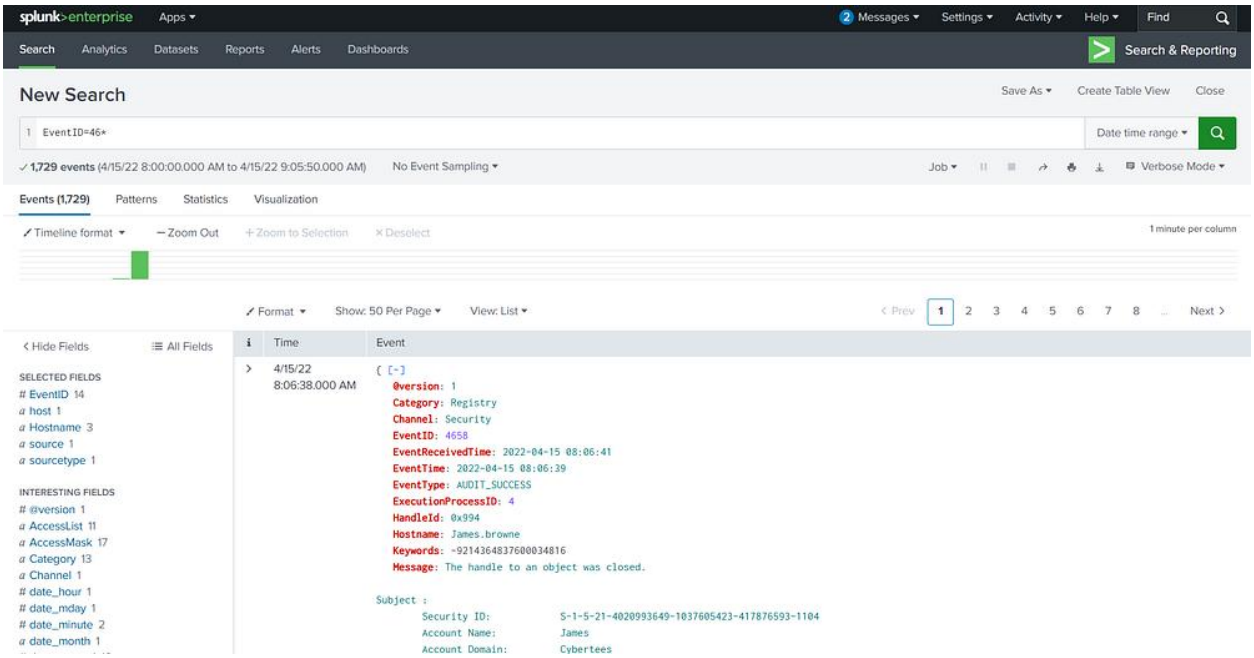
Splunk supports wildcards and CIDR search for IP addresses.

*(Asterisk): Represents zero or more

Example:

```
EventID=46*
```

```
DestinationIp=172.16.0.0/24
```

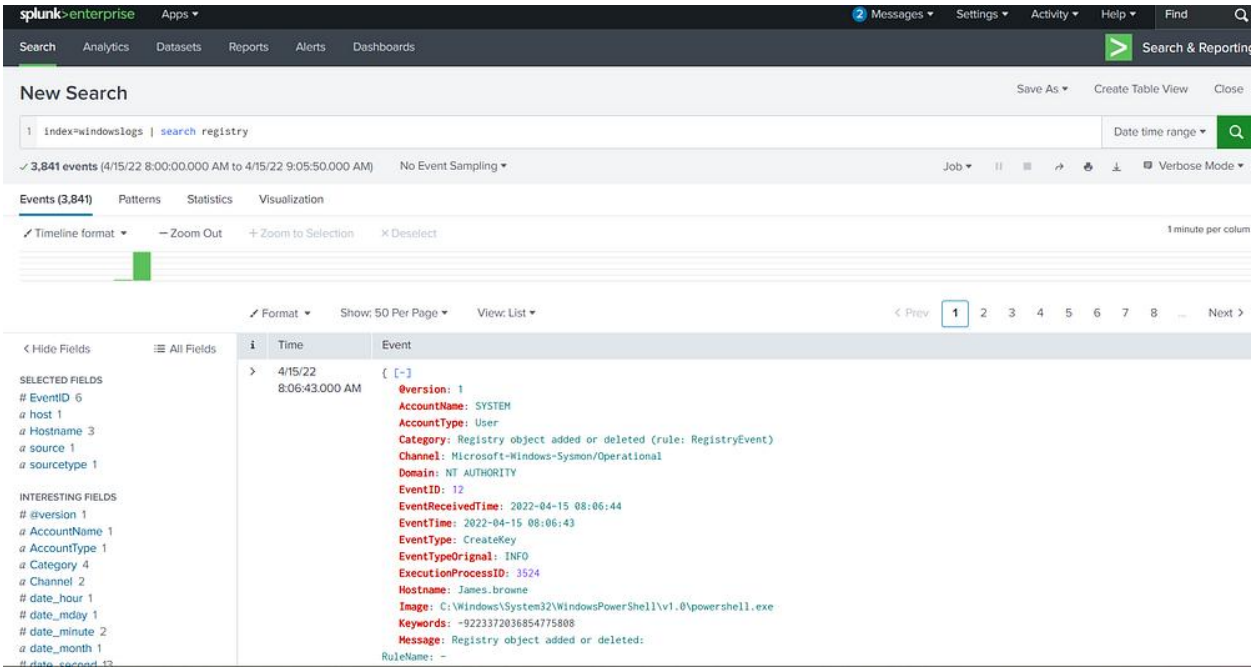


4. Pipe Operator (|)

Passes the output of one command into the next command.

Example:

```
index=windowslogs | search registry
```



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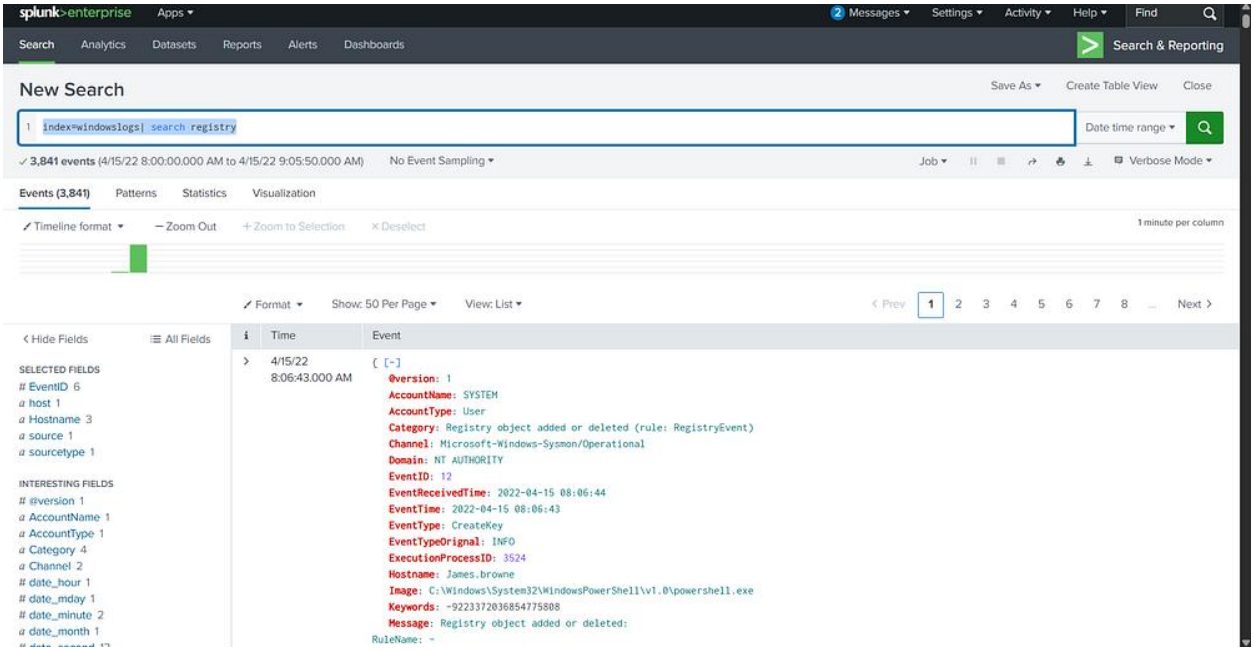
5. Search

This command is used for the raw text while using the pipe operator (|).

Example:

```
index=windowslogs | search registry
```

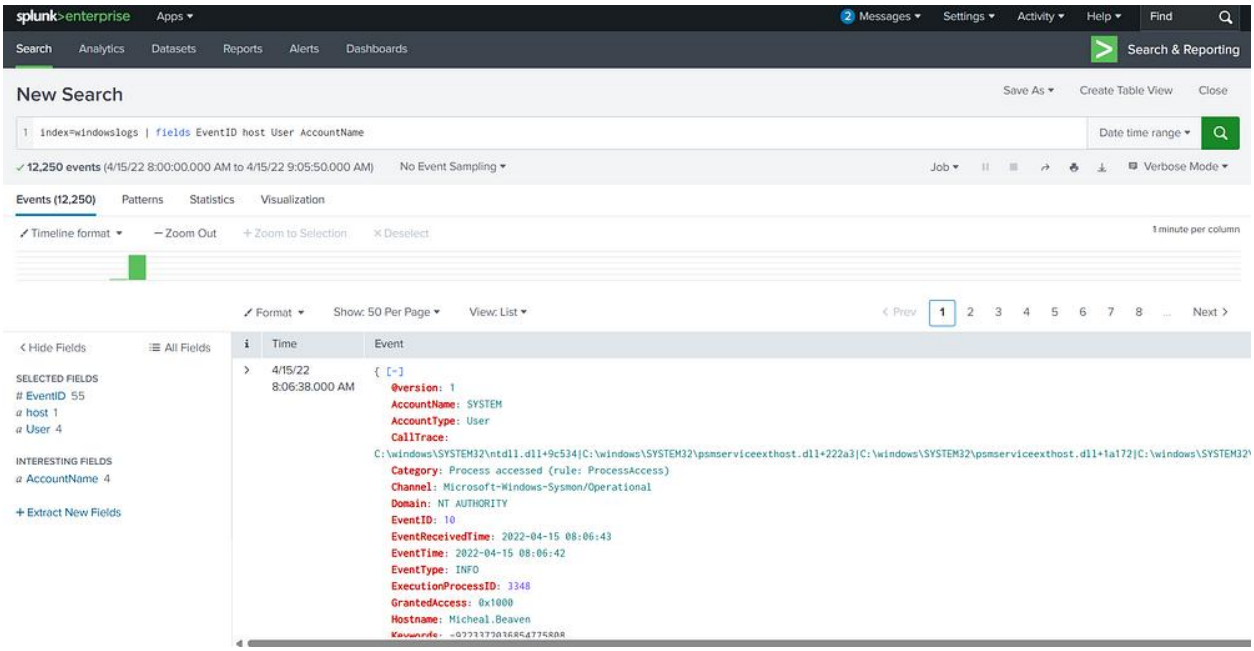


6. Fields

Fields commands is used to add or remove mentioned fields from the search results. To remove the field from the search result minus sign (-) is used and plus (+) sign is used before the field name which we want to display. If no operator is specified before the field name then its plus(+) by default.

Example:

```
index=windowslogs | fields +EventID +host +User +AccountName -SourceIp
```

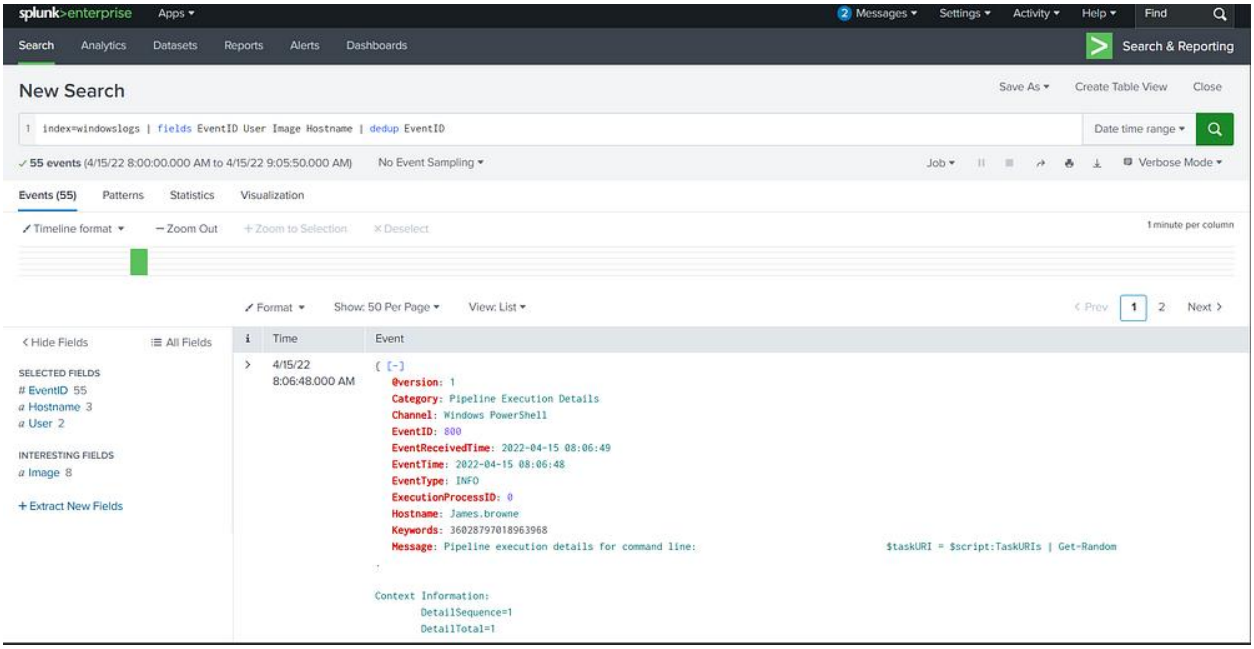


7. Dedup

Dedup is the command used to remove duplicate fields from the search result.

Example:

```
index=windowslogs | fields EventID User Image Hostname | dedup EventID
```



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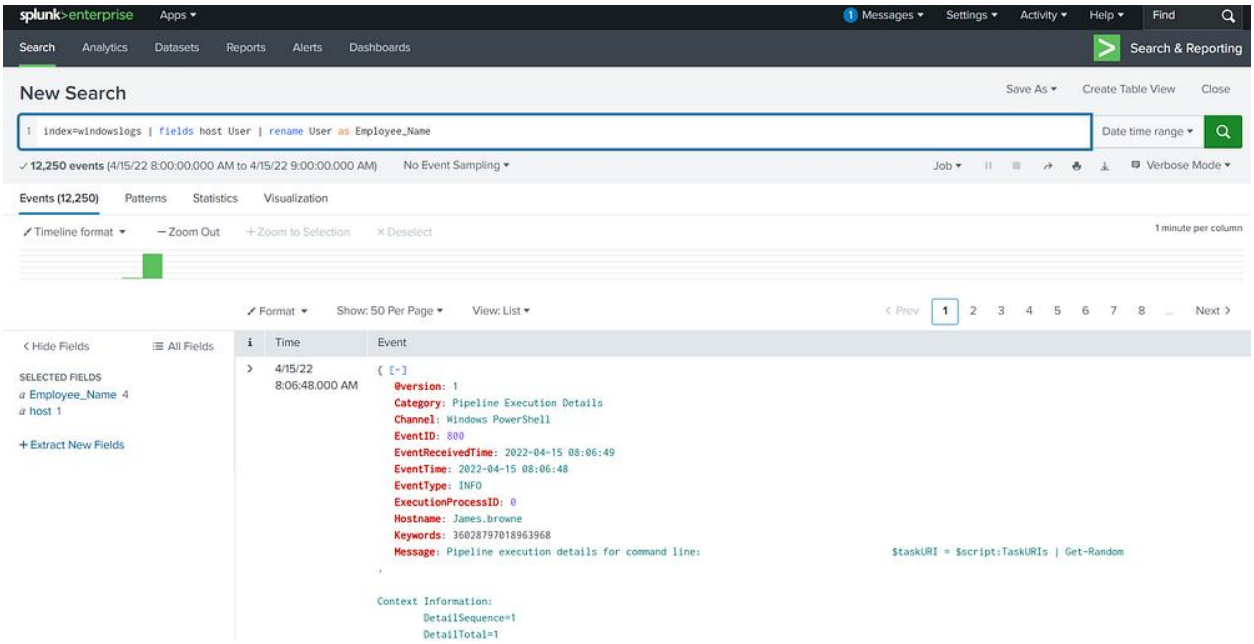
8. Rename

It allows us to change the name of the field in the search result.

Example:

```
index=windowslogs | fields host User | rename User as Employee_Name
```

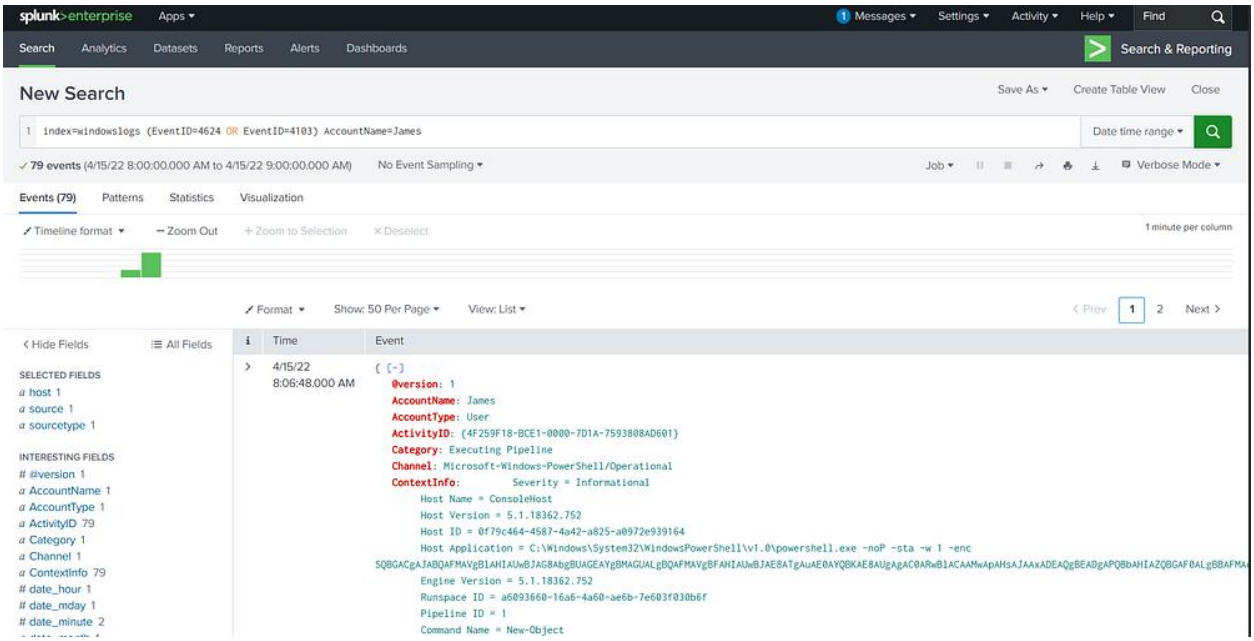


9. Parentheses ()

Used for grouping logic.

Example:

```
index=windowslogs (EventCode=4624 OR EventCode=4103) AccountName=James
```

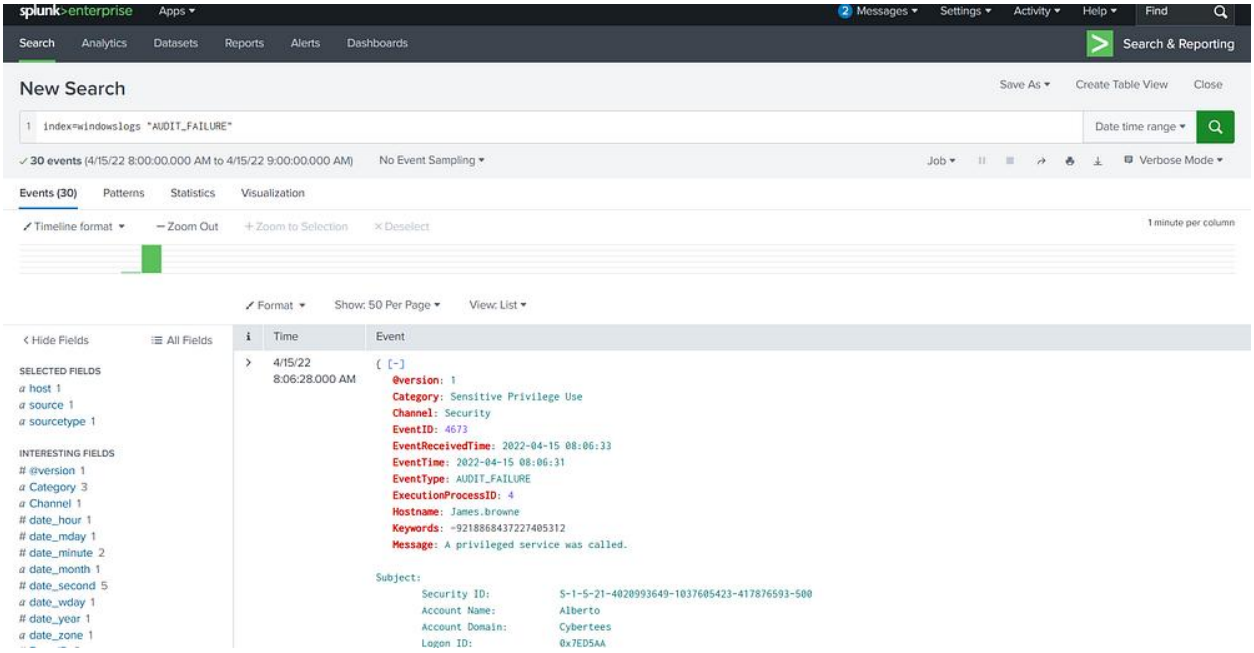


10. Quote Operator “”

Used for exact phrase match.

Example:

```
index=windowslogs "AUDIT_FAILURE"
```

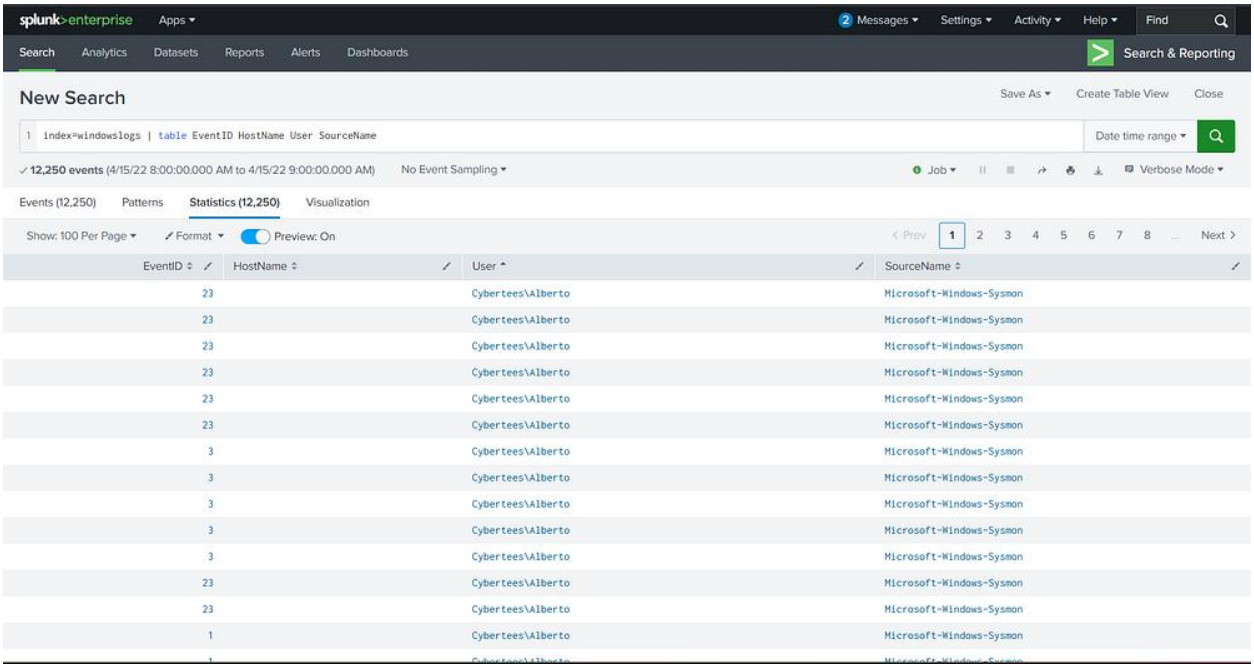


11. Table

The table command us to create a table with selective fields as columns.

Example:

```
index=windowslogs | table EventID HostName User SourceName
```

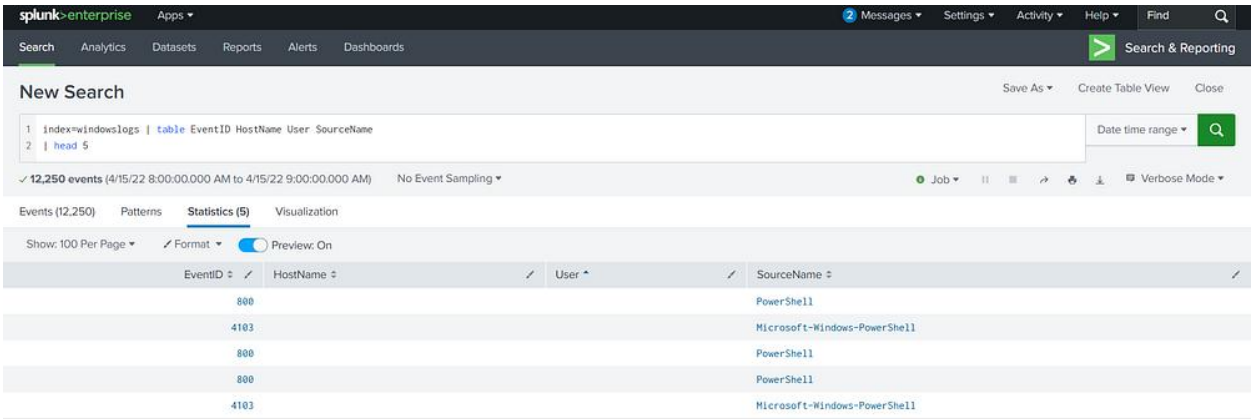


12. Head

The head command returns the first 10 events if no number is specified.

Example:

```
index=windowslogs | table EventID HostName User SourceName| head 5
```

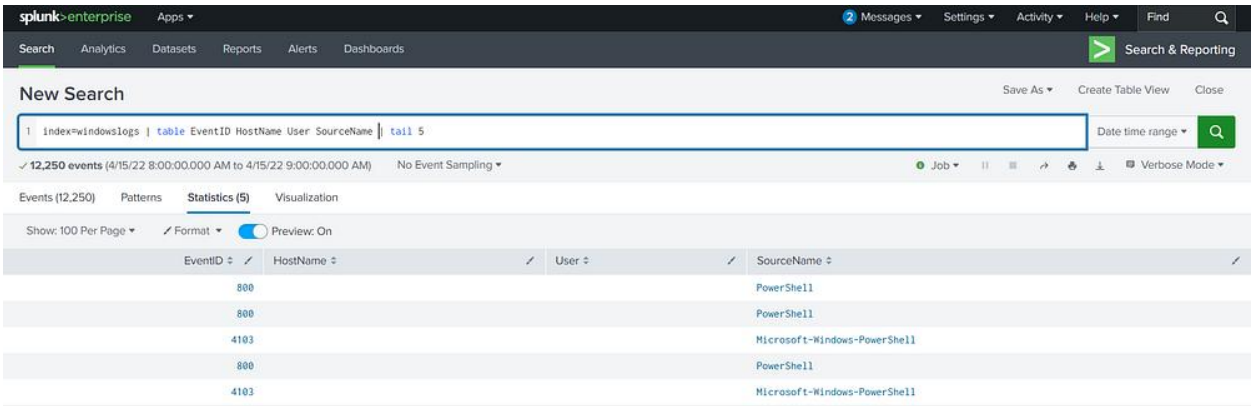


13. Tail

The tail command returns the last 10 events if no number is specified.

Example:

```
index=windowslogs | table EventID HostName User SourceName | tail 5
```

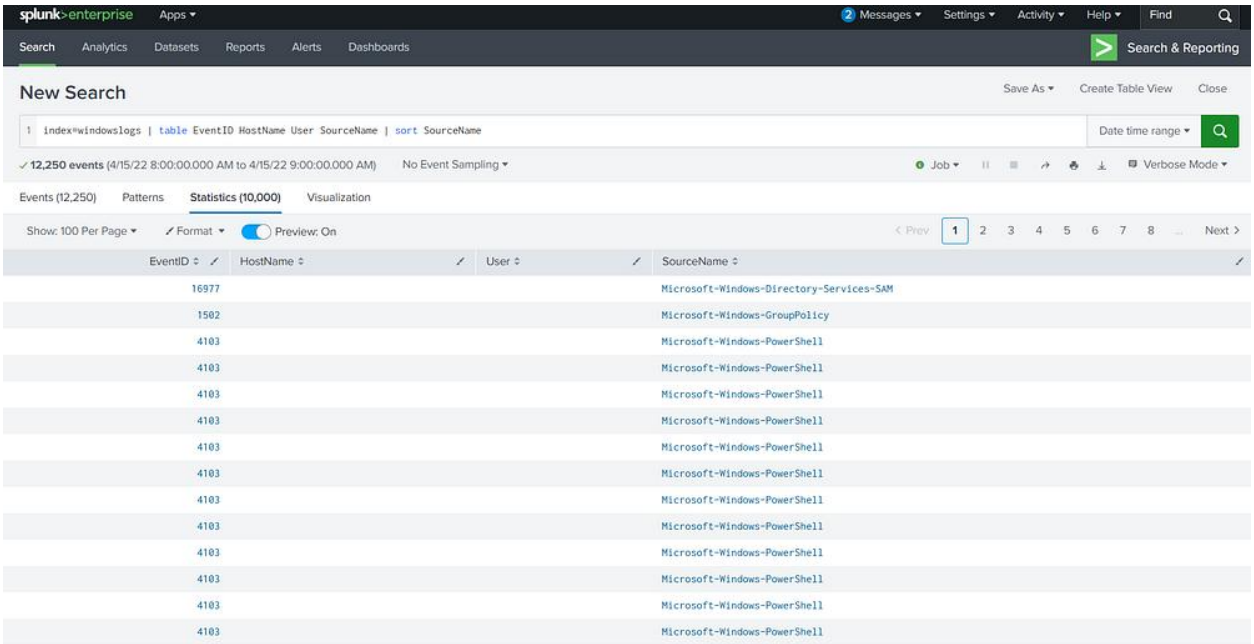


14. Sort

The sort command allows us to order the fields in ascending or descending order.

Example:

```
index=windowslogs | table EventID HostName User SourceName | sort SourceName
```

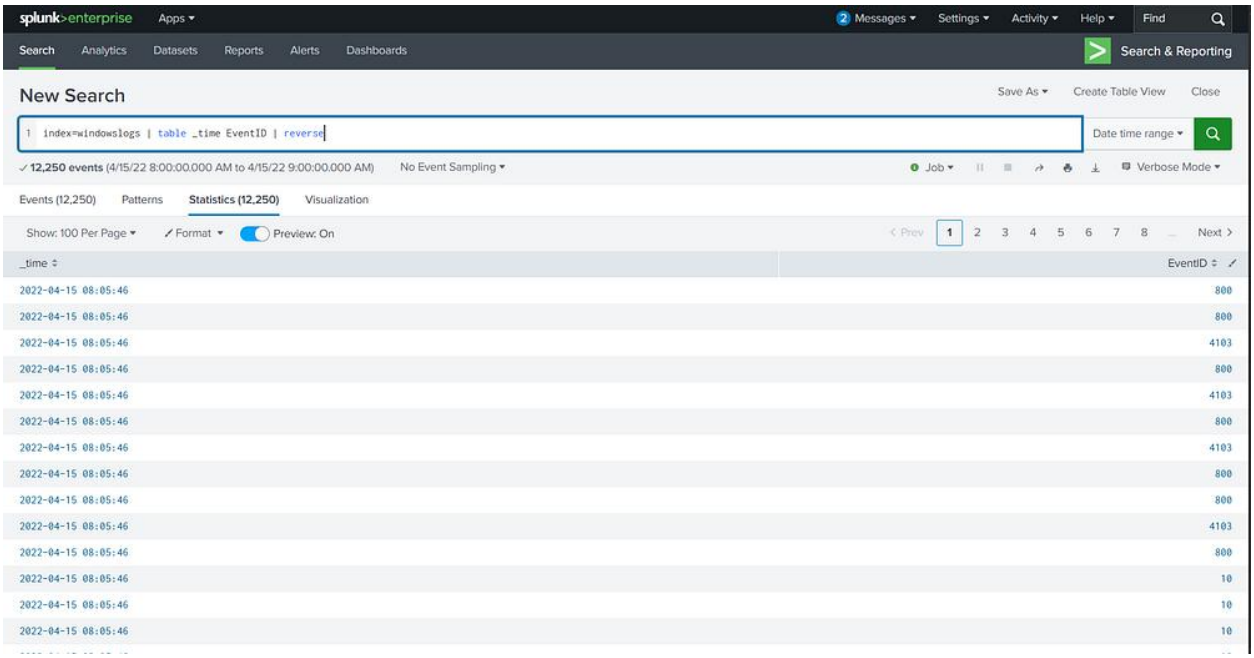


15. Reverse

The reverse command simply reverses the order of the events.

Example:

```
index=windowslogs | table _time EventID | reverse
```

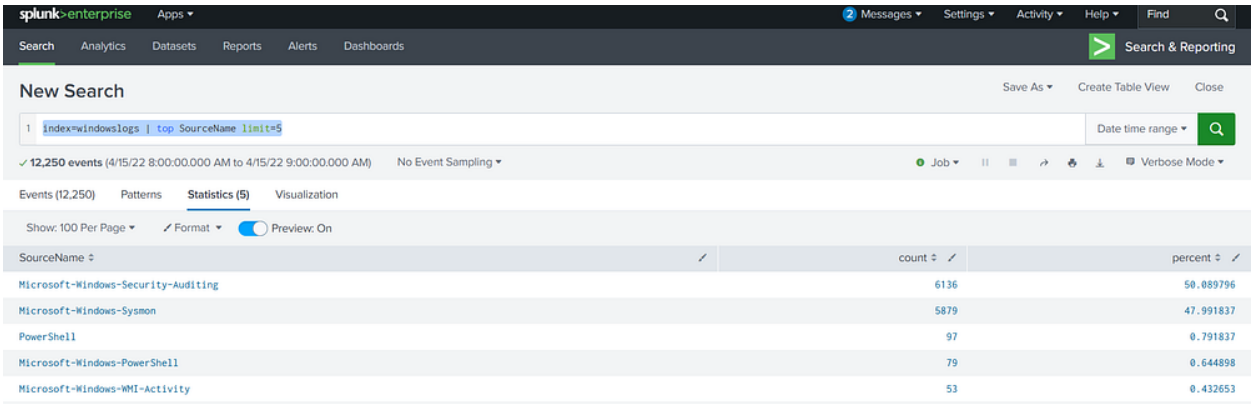


16. Top

This command returns the most frequent values for the field specified.

Example:

```
index=windowslogs | top SourceName limit=5
```

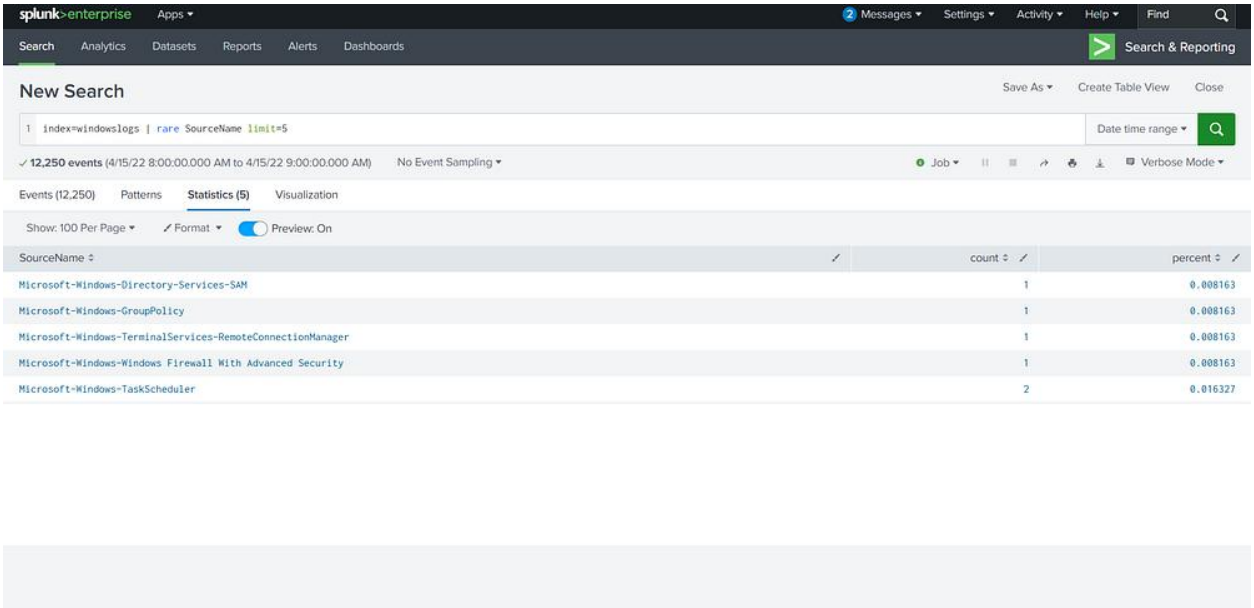


17. Rare

This command does opposite of the top command, as it returns least frequent values.

Example:

```
index=windowslogs | rare SourceName limit=5
```

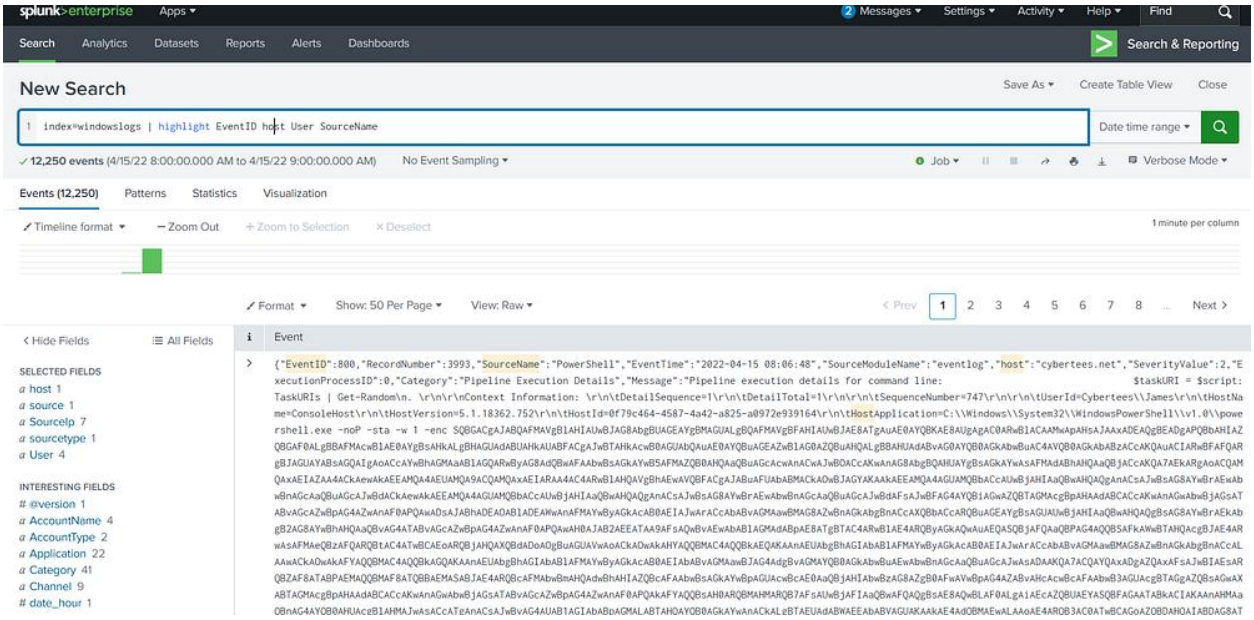


18. Highlight

The highlight command shows the results in rqw events mode with fields highlighted.

Example:

```
index=windowslogs | highlight EventID host User SourceName
```



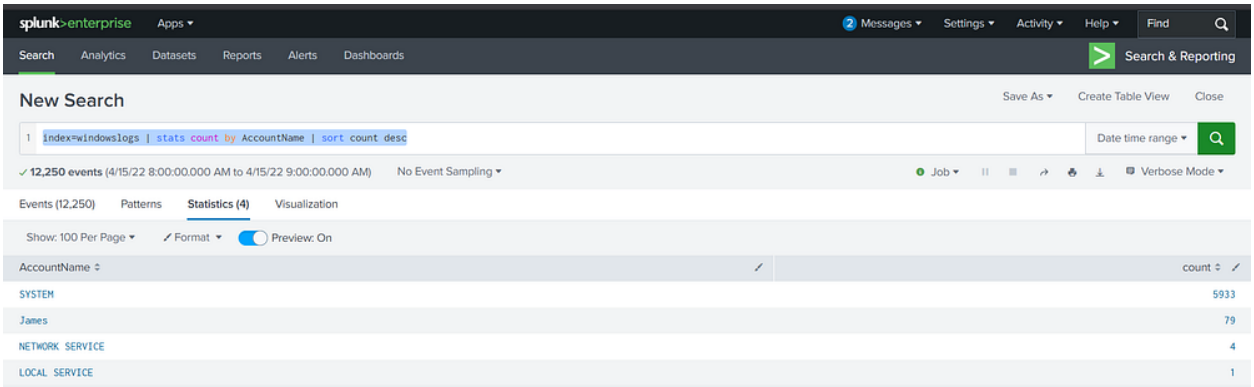
19. STATS commands

SPL supports various STAS commands that help in calculating statistics on the values. Some common STAS commands are:

- stas avg(file_name)
- stas max(file_name)
- stas min(file_name)
- stas sum(file_name)
- stas count by SourceIp

Example:

```
index=windowslogs | stats count by AccountName | sort count desc
```

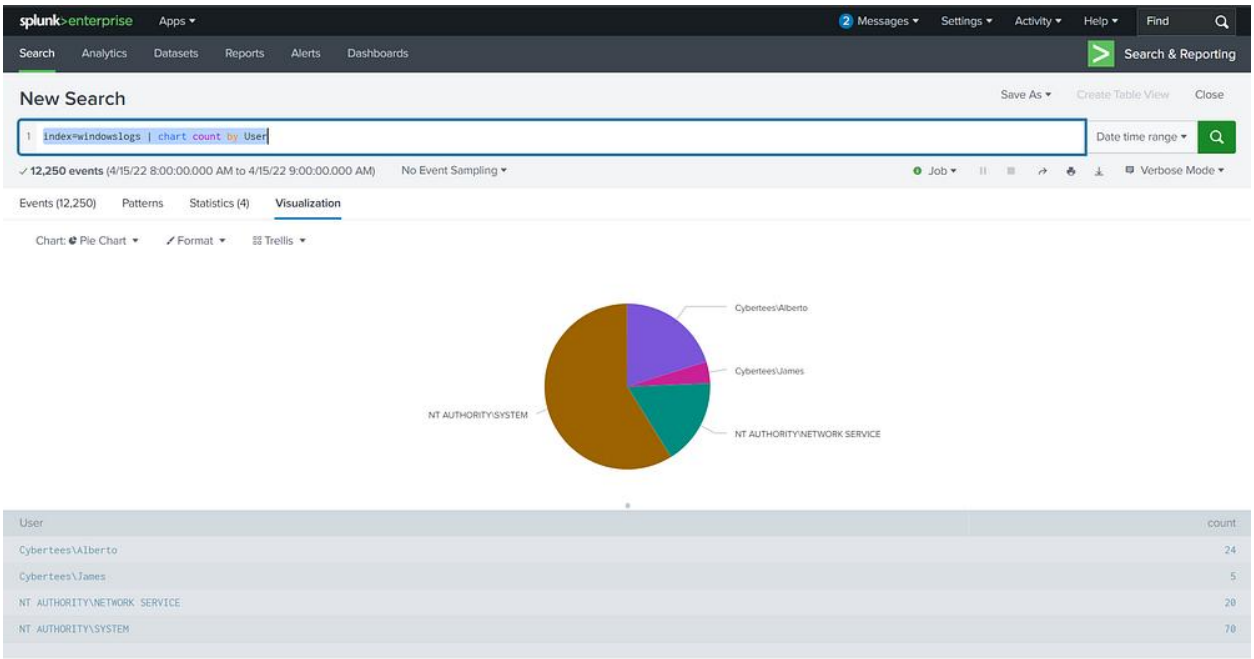


20. Chart

Chart command is used to transform the data into tables or visualizations.

Example:

```
index=windowslogs | chart count by User
```

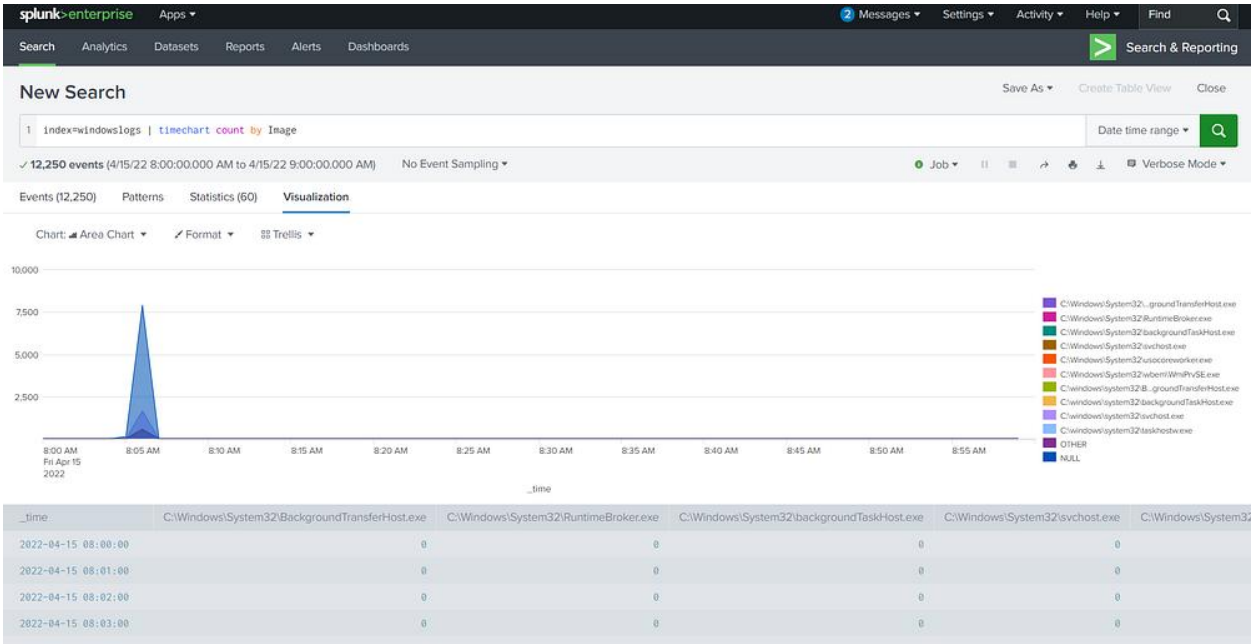


21. Timechart

Timechart command returns the time series chart covering the field following the function mentioned. Often combined with STAS command.

Example:

```
index=windowslogs | timechart count by Image
```



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